# Why Should This Work?

## Once the backtest is honest, what is left is economics

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# Monday: the record you are shown was selected

- The managers who come to see you are the ones whose record cleared the bar
- Journals work the same way: they print the strategies whose t-statistic cleared the bar

Today: the published record, and what is left to ask once the backtest is honest

# Today's question

Why should anyone pay you a premium for holding this?

- Statistics can tell you a strategy made money. It cannot tell you why it should keep making it.
- That takes a story: **who is on the other side of the trade, and why they are willing**

# Predict first: how many survive?

27 of our strategies were published with a significant t-statistic. Re-run on our sample, 1980–2000:

How many clear the Bonferroni bar, $|t| > 3.11$?

**A** 0 to 6

**B** 7 to 13

**C** 14 to 20

**D** 21 to 27

Write your answer down. You will type it into the notebook, and the notebook will tell you.

# Predict first: three papers, how many trades?

Three volatility papers, two sets of authors: idiosyncratic volatility and realized volatility (Ang et al. 2006), and lottery-like stocks (Bali, Cakici and Whitelaw 2011).

How correlated are their long-short returns?

**A** below 0.3

**B** 0.3 to 0.6

**C** 0.6 to 0.9

**D** above 0.9

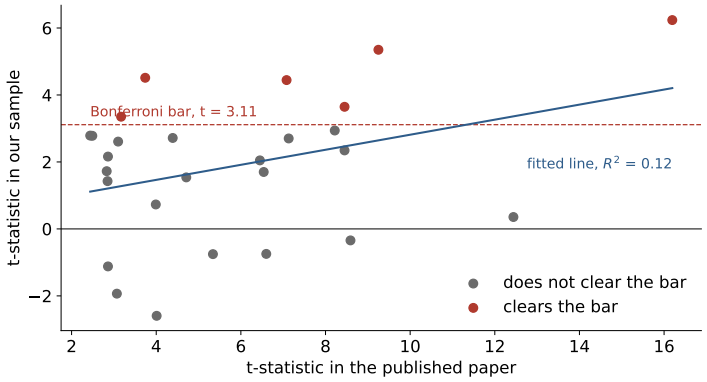
# Now: the notebook

**Notebook: Why Should This Work? (in class)** — about 30 minutes, in pairs

1. **§1 The gap.** Type your first prediction, then run the replication table.
2. **§2 Count ideas, not papers.** Type your second prediction, then run the volatility family.
3. **§3 Prompt it.** *"What was the worst drawdown?"* Write the prompt, then run the check and read the code.
4. **§4 Your strategy.** Its closest neighbours, and its worst episode: one line to write.

**Note:** cells marked with a lock run as they are. Stop at **Back to the slides**; everything after it (§5 onwards) is for after class.

# Six of 27 survive




- Six come out with the **wrong sign**. The published t explains 12% of ours.
- Knowing a paper reported $t = 9$ rather than $t = 3$ tells you almost nothing about what you will get

# Three explanations, one table

| | mechanism | so the effect . . . |
|---|---|---|
| **Selection** | journals print what worked; the failures stay in a drawer | never existed |
| **Overfitting** | the authors searched too | never existed |
| **Decay** | it was real; publication revealed it; investors traded it away | existed, and is gone |

A replication gap cannot tell you which. Two say never trade it; one says you are late.

# Risk for whom?

- Risk is not a property of an asset. It is a property of an asset **for a particular investor**.
- You are paid to hold something that **pays off badly in bad times**, when you can least afford it
- The CAPM is one narrow version: it declares that bad times means the market falling
- But bad times are about people: a lost job, a house that falls with the market, a credit line that disappears

# Differ from the market only if you differ

You work in the auto industry. Your income already moves with autos, so hold **the market minus autos**.

| | | what you should do |
|---|---|---|
| **You are the average investor** | your bad times are the market's | hold the market |
| **You genuinely differ** | what scares them does not scare you | take the other side, and get paid |
| **You think you differ, and do not** | the most common case | you find out at the worst time |

- Two investors can agree on every statistic and still correctly reach opposite decisions

# As you scale, you become the investor bearing the risk

- "Risky for them, not for me" is a statement about a **small** position
- Put enough capital behind it, and you are the one holding through the drawdown and meeting the margin call
- If everyone held it, prices would adjust and the premium would shrink

Can the world support everyone picking up this $100 bill?

# A high expected return is a low price

"These stocks earn more" is the same sentence as **"these stocks are cheap"**. Someone is selling to you at that price.

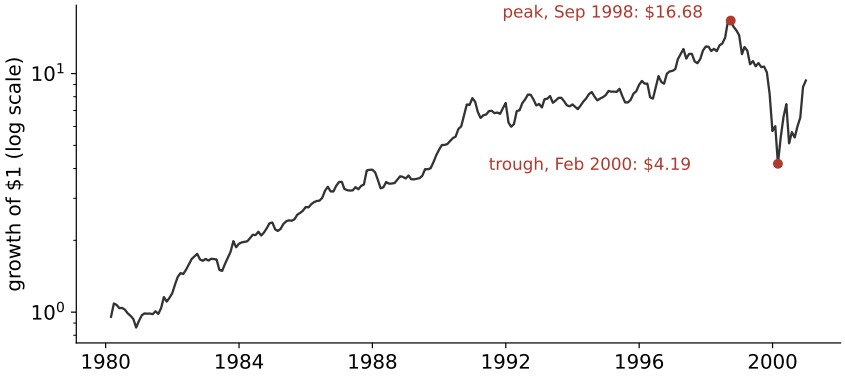
| | why the price is low | what you are buying |
|---|---|---|
| **Cash flows** | the market expects the business to do badly, and is right | a bad business at a fair price |
| **Risk** | it pays off badly when investors care most | compensation for bearing something real |

Who is selling to you, and what makes them willing? If you cannot name them, you are the one being selected.

# Low volatility: who is on the other side?

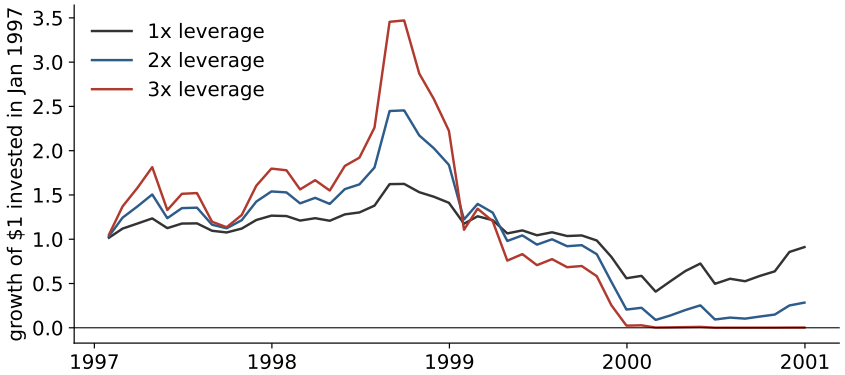
- **The claim:** quiet, low-beta stocks earn more than their risk justifies
- **The counterparty:** investors who want equity-like returns but cannot borrow (mandates, regulation, margin). To reach for return, they buy risky stocks.
- That bids up the volatile names and leaves the quiet ones cheap
- Their reason is a constraint, not a mistake. Constraints do not go away because someone points them out.
- Is it risky for *you*? Only if you are unconstrained: if you can borrow, and if you can hold.

# What holding it felt like




- 15.4% a year from 1980 to 1997. Then 1999: −60%.
- The strategy is short lottery stocks, and in a bubble lottery stocks are exactly what runs

# Leverage turned a bad year into a wipeout




- At $2\times$ you lose 96%. At $3\times$ you are wiped out in June 2000, before the recovery.
- Harvesting this premium needed leverage. The risk that was other people's became yours when you scaled.

# Overfitting, or the premium being paid?

- In the backtesting lecture this was the in-sample winner: Sharpe ratio **1.37** in 1980–1990, **0.22** in 1991–2000
- **If it is overfitting:** drop it
- **If it is the premium being paid for:** the bad state the story predicts arrived. Size it to survive the next 1999.

Nothing in the returns tells you which. Only a story written down beforehand does.

# Worked problem: what was the worst drawdown?

A long-short strategy earns these returns over four months:

| month | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| return | $+10\%$ | $-20\%$ | $-20\%$ | $+5\%$ |

(a) Worst drawdown, funded 1-for-1 with equity, compounded

(b) The same at $2\times$ leverage

(c) The same, summing the returns instead of compounding

Work it by hand. The notebook's §5 has a cell to check it.

# Worked problem: three answers from the same returns

| value after month | 1 | 2 | 3 | 4 | worst drawdown |
|---|---|---|---|---|---|
| (a) $1\times$, compounded | 1.10 | 0.88 | 0.704 | 0.739 | $0.704/1.10 - 1 = -36\%$ |
| (b) $2\times$, compounded | 1.20 | 0.72 | 0.432 | 0.475 | $0.432/1.20 - 1 = -64\%$ |
| (c) summing returns | $+10$ | $-10$ | $-30$ | $-25$ | $-30 - 10 = -40$ points |

- A long-short has no capital of its own. A drawdown is undefined until you say what capital you posted.
- Leverage deepens the same loss: $-36\%$ at $1\times$, $-64\%$ at $2\times$, $-84\%$ at $3\times$

# Five questions before you trade a signal

1. **Who is on the other side**, and what makes them willing?
2. Is their reason a **constraint, a preference, or an error**? Constraints persist; errors get arbitraged.
3. Is the risk one **you** can bear, at the size you intend to trade?
4. What would you have to see to stop believing? **Name it now**, while it is cheap.
5. If the answer to (1) is "I don't know", **you are the counterparty**

# Homework: the valuation family

Do to value what we did to volatility: book-to-market (`BM`), earnings-to-price (`EP`) and the enterprise multiple (`EntMult`). Three papers, three decades.

- Q1 **How many ideas is this?** The mean pairwise correlation of the three
- Q2 **The worst episode.** The composite's drawdown, funded 1-for-1, compounded, and when it happened
- Q3 **Is it a hedge?** The composite's market beta
- Q4 **The memo.** How many bets is this, and who is on the other side of the value trade: a constraint, a preference, or an error?

Same steps as today's volatility family. It is at the end of today's notebook, after §5. Due before the next class.

# Takeaways

- A replication gap has three explanations, and the data cannot separate them
- Count ideas, not papers: three volatility papers are one trade
- Risk is a property of an asset **for an investor**. Deviate from the market only if you differ.
- A high expected return is a low price: ask who is selling, and why
- As you scale, you become the investor bearing the risk
- A drawdown is undefined for a long-short until you say what capital you posted